Lab 7: Fuzzing for Input Validation Bugs

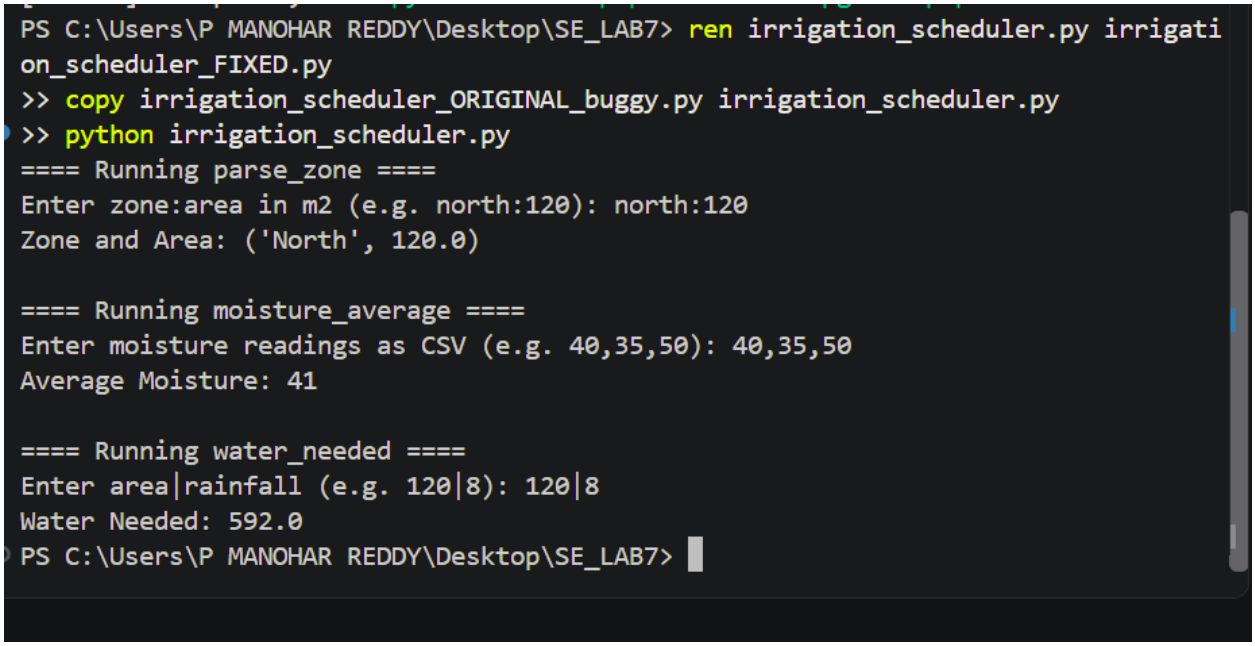
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Project: Irrigation Scheduler (irrigation_scheduler.py)

Bugs found by fuzzing

parse_zone: crashed on None, strings without ':' , and ':' (empty area). moisture_average: crashed on None and empty/non-numeric items such as '' (also divided by zero on empty list). water_needed: crashed on None, strings without '|', and '|' (empty values).
Fix: each function now checks the input type, validates the format, catches conversion errors, rejects nan/inf/negative values, and returns None for invalid input. main() prints 'Invalid input' in that case.

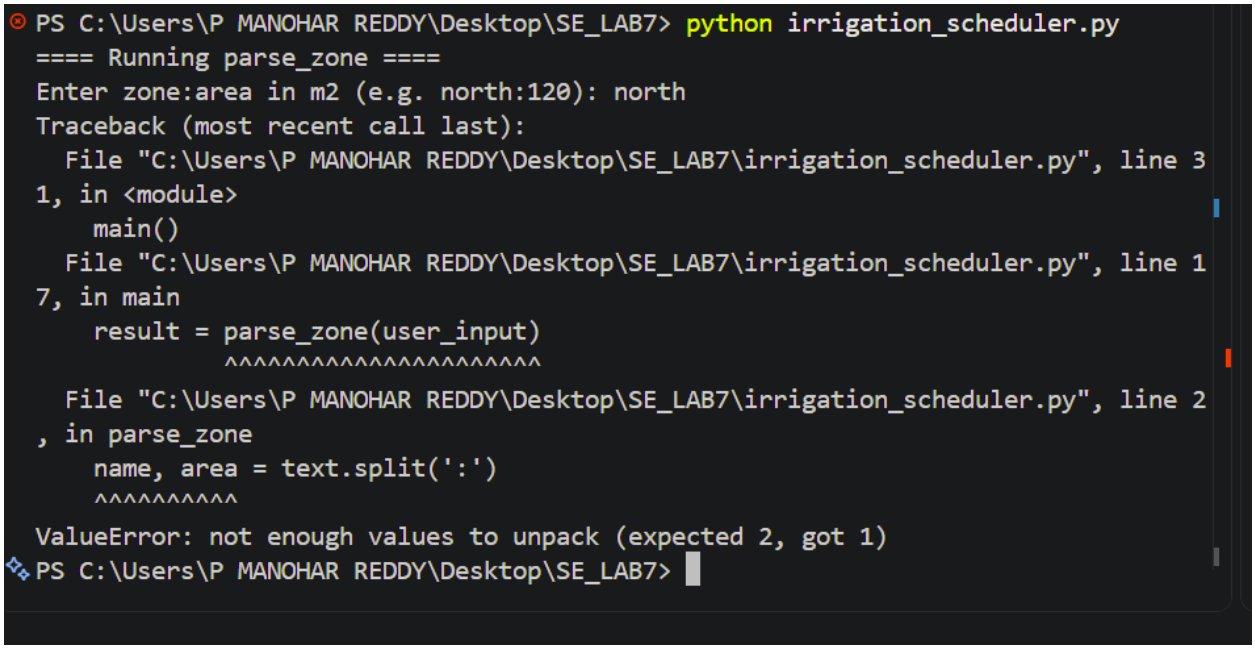
Screenshot 1: Step 2 - Buggy application: normal input

Running python irrigation_scheduler.py (original, buggy version) with valid input works as expected.



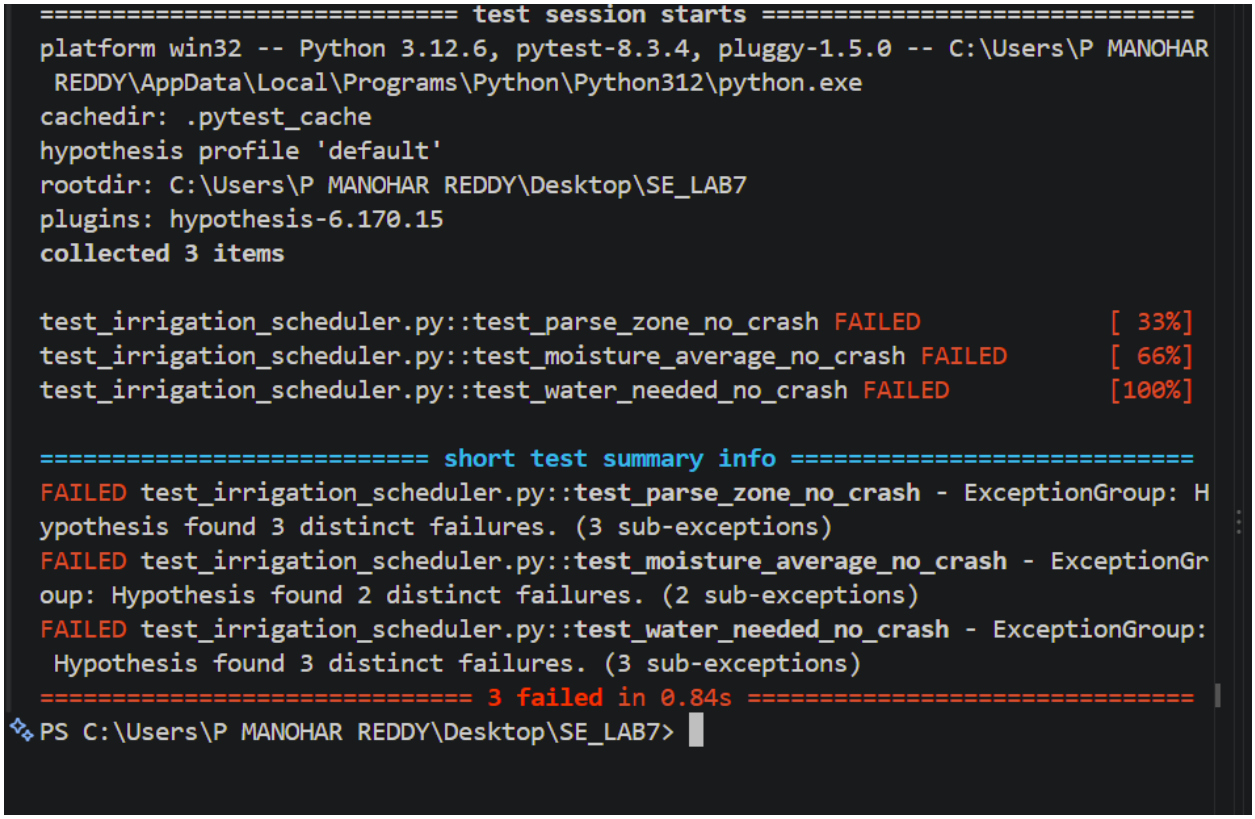
Screenshot 2: Step 2 - Buggy application: crash on invalid input

The buggy version crashes with ValueError when the zone input has no ':' separator.



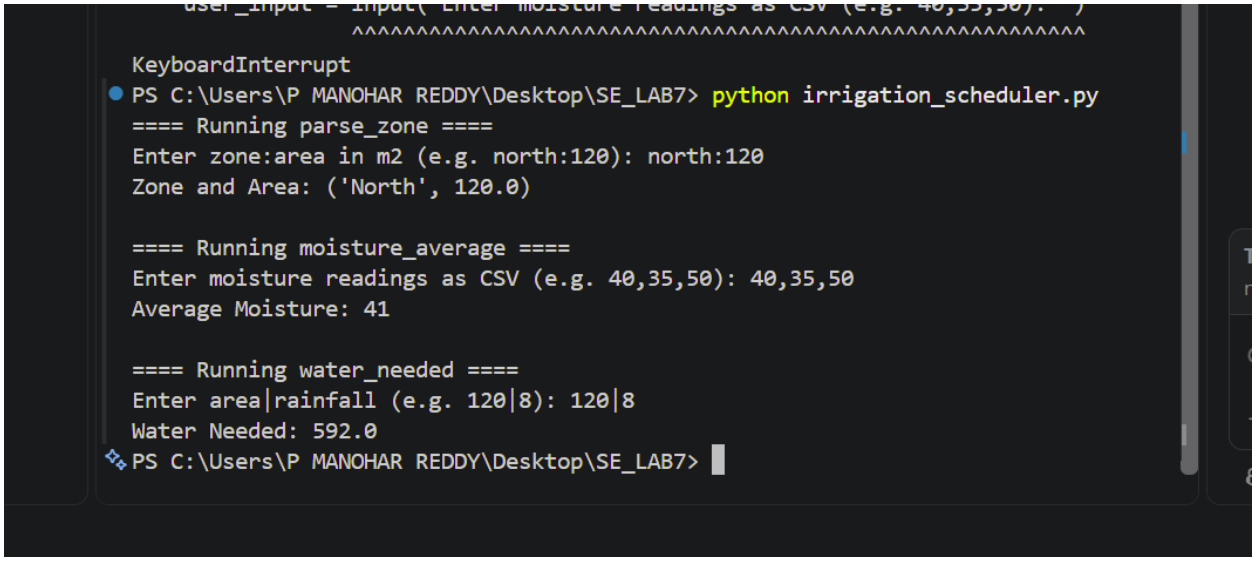
Screenshot 3: Step 4 - Hypothesis fuzz tests failing on buggy code

All 3 Hypothesis tests fail. Hypothesis found crashing inputs such as None, '', ':' and '|'.



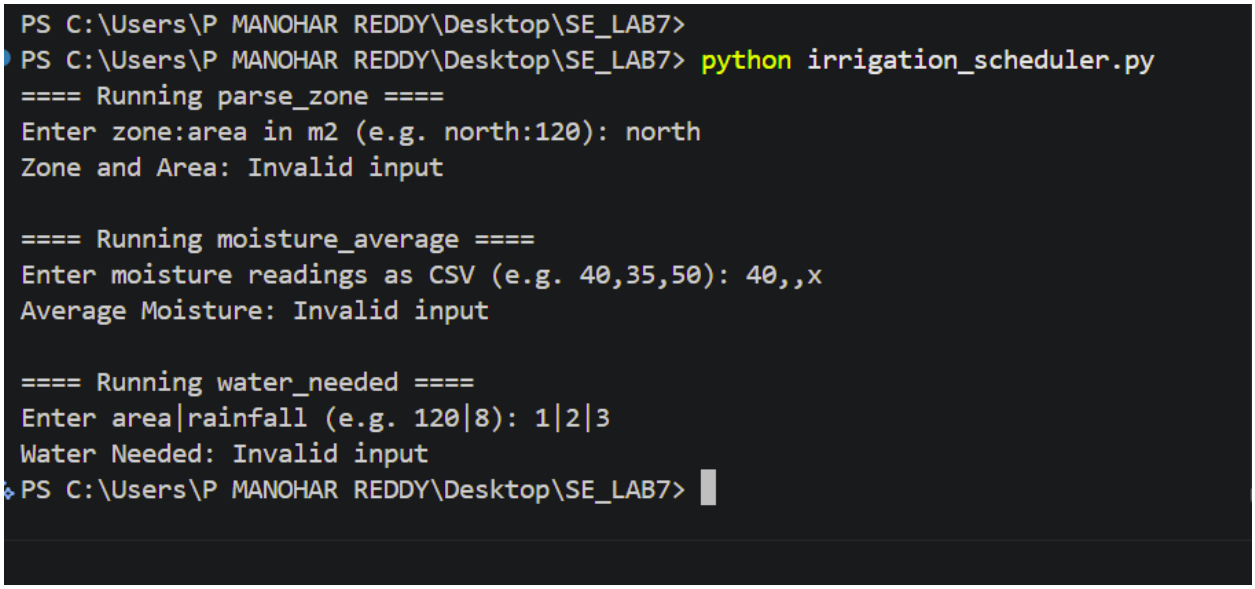
Screenshot 4: Step 5 - Fixed application: normal input

Running the fixed irrigation_scheduler.py with valid input gives the same correct results.



Screenshot 5: Step 5 - Fixed application: invalid input handled

The fixed version handles invalid inputs gracefully and prints 'Invalid input' instead of crashing.



Screenshot 6: Step 6 - Hypothesis fuzz tests passing on fixed code

All 3 Hypothesis tests pass against the fixed irrigation_scheduler.py.

